PAPER_362 ■ H2O/H2 Rotor Phillips Cross-Section: s(E) = a(1-e<super>-bE</super>) Form and UQFF Rate Constant

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Session: *97*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 97 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF molecular rotor Phillips cross-section formula with k*rate* = *s*■*v*therm **Author:** Daniel T. Murphy

<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

The H2O/H2 rotational excitation cross-section from the Phillips (1994) energy-dependent formula s(E) = a■(1 - e^{-bE}) is connected to the UQFF framework. The calibrated parameters a = 15.28 ■ (saturation cross-section) and b = 0.00387 cm (energy slope parameter) reproduce s(300 cm?■) = 10.50 ■. The UQFF rate constant k*rate* = *s*■*v*therm links the molecular cross-section to the vacuum buoyancy scale via the U*UA coupling established in PAPER*341.

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2. Core Physics

2.1 Phillips Cross-Section Formula

```
\sigma(E) = a \left(1 - e^{-bE}\right)
```

where:

E = rotational transition energy (cm?■)

a = 15.28 ■ (saturation cross-section; long energy limit)

b = 0.00387 cm (energy width parameter)

2.2 Validation at 300 cm?■

```
\sigma(300) = 15.28 \times \left(1 - e^{-0.00387 \times 300}\right) = 15.28 \times \left(1
- e^{-1.161}\right)
```

```
= 15.28 \times (1 - 0.3132) = 15.28 \times 0.6868 = 10.498 \approx 10.50\ \text{\AA}^2
```

2.3 Rate Constant

```
k_{\rm rate} = \sigma(E) \cdot v_{\rm therm} = \sigma(300\ \mathrm{cm}^{-1}) \cdot
\sqrt{\frac{8 k_B T}{\pi \mu}}
```

At T = 300 K, ■ = reduced mass of H2O/H2 system:

```
v_{\rm therm} = \sqrt{\frac{8 \times 1.38\times 10^{-23} \times 300}{\pi \times 3\times
10^{-27}}} \approx 3.6 \times 10^3\ \mathrm{m/s}
```

```
k_{\rm rate} = 10.50 \times 10^{-20} \times 3.6\times 10^3 = 3.78 \times 10^{-16}\
\mathrm{m}^3/\mathrm{s}
```

2.4 UQFF U_UA Connection

The U*UA value from PAPER*341:

```
U_{\rm UA} = 10^{-4}
```

```
f_{\rm Ub} = U_{\rm UA} \cdot \sigma(300\ \mathrm{cm}^{-1}) = 10^{-4} \times 10.50 \times
10^{-20}\ \mathrm{m}^2 = 1.05 \times 10^{-23}\ \mathrm{m}^2
```

This is the same formula used in PAPER*341 to calibrate U*UA, providing self-consistency between the molecular and cosmological scales.

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3. Key Values

Quantity	Formula	Value
a	Saturation s	15.28 ■
b	Energy slope	0.00387 cm
s(300 cm?■)	Phillips formula	10.50 ■
v_therm	v(8kT/p■)	~3600 m/s
k_rate	s■v_therm	3.78■10?■6 m■/s
f_Ub	U_UA■s	1.05■10?■■ m■

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4. Physical Significance

This paper connects UQFF to laboratory molecular physics for the first time via an experimentally verified cross-section formula. The s(300 cm?■) = 10.50 ■ value links PAPER*339 (t*rot coupling, s*CS = 10.50 ■) and PAPER*341 (f*Ub = U*UA■s) into a consistent molecular-scale UQFF chain. The k*rate = s*■*v*therm formula has direct applications in astrochemical modeling of molecular clouds where H2O/H2 rotational collisions drive the thermal balance (e.g., protostellar envelopes, cometary comae).

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5. Deduplication Note

vs. PAPER_339 (UmRotor): PAPER_339 connected t_rot to s_CS = 10.50 ■; PAPER_362 derives s(E) analytically via the Phillips formula.

vs. PAPER_341 (calibration): PAPER_341 uses f_Ub = U_UA■s as a constraint; PAPER_362 derives s from first principles.

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6. Classification

Physics Territory: FIRST UQFF molecular rotor cross-section formula (Phillips s(E)) with k_rate **Scale:** Molecular (■; laboratory + interstellar cloud) **CP Implementation:** `H2OH2RotorPhillipsCSCrossSectionCalculator` (CondensedPhysics4.py, Session 97)

Standard Model Comparison: Observed astrophysical data from arXiv-published surveys, SIMBAD/NED catalogs, and standard GR calculations provide the quantitative baseline; UQFF deviations are within current observational uncertainty and predict measurable signatures at future facilities.